

Lecture

jQuery

What is a library?

— — —

How would you describe a library to someone who has never encountered the word?



Students, write your response!

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What is a library?

A "library" is a collection of program parts that do common and/or specialized things that save the programmer from needing to "reinvent the wheel"

Most libraries consist of functions to call:

```
var randomNum = Math.floor(Math.random() * 6);
```

Can you think of any other functions built into javascript?

Math is a library that is built into JavaScript. It has [26 functions](#) for performing math-related calculations!

Most other libraries need to be *imported*.

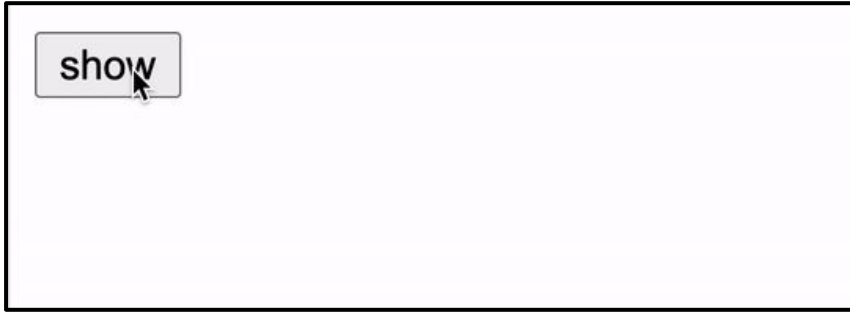


Students, write your response!

jQuery

— — —

The **jQuery** library provides functions for manipulating the HTML elements and CSS styles of a website using JavaScript code.



According to [builtWith](#), almost 90 million websites use jQuery!

Try it out!



Students browse: trends.builtwith.com/javascript/jquery

jQuery Syntax

— — —

All jQuery code involves

1. interacting with an **HTML element**
2. performing an **action** using/on/with that element.

```
$( "selector" ).action( args )
```



```
// tell the button to listen for clicks and
// to react with toggleSecret()
$( "button" ).on( "click", toggleSecret );

// function for reacting to button clicks
function toggleSecret() {
    if ( $( "button" ).text() === "show" ) {
        $( "p" ).show();
        $( "button" ).text( "hide" );
    }
    else {
        $( "p" ).hide();
        $( "button" ).text( "show" );
    }
}
```

jQuery Syntax: Different Example

All jQuery code involves

1. interacting with an **HTML element**
2. performing an **action** using/on/with that element.

```
$( "selector" ).action( args )
```

```
$( "#chaos" ).on( "click", changeStuff );

// function for reacting to "chaos" clicks
function changeStuff() {
    $( "button" ).text( "don't click" );
    $( ".secrets" ).show();
    $( "p" ).addClass( "magic" );
    $( "#melon" ).css( "color", "green" );
    $( ".magic" ).text( "abra kadabra" );
}
```

Code Along - jquery cdn

Import jQuery by adding the tag below inside the <head>

```
<script src="https://code.jquery.com/jquery-git.js">
```

<https://code.jquery.com/jquery-git.js>

Keep your JSBin open, you'll submit the link in a little while.

Right below the closing body tag, add script tags:

```
<body>
  <div id='myID'>
    <h1 class="italic"> Hello World</h1>
    <p class="italic"> jQuery is Awesome!</p>
    <p> we can do so much </p>
    <p> like flip a coin: </p>
    <p id='result'> pending </p>
  </div>
</body>

<!-- Add a pair of <script></script> tags here-->
```

Manipulate the HTML by inserting the code on the right into the new script tags one line at a time

```
<div id='myID'>
  <h1 class="italic"> Hello World</h1>
  <p class="italic"> jQuery is Awesome!</p>
  <p> we can do so much </p>
  <p> like flip a coin: </p>
  <p id='result'> pending </p>
</div>
```

```
// select by element type
$("h1").text("Flipadeedoodah");
$("p").removeClass("italic");

// select by class
$(".italic").css("color", "red");

// select by id
$("#result").addClass("italic");
$("#myID").empty();
```

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Students, write your response!

We want the #result paragraph element to randomly show either "heads" or "tails" each time the page reloads.

```
<html>
  <head>
    <!-- Load in the jQuery Library-->
    <script src="https://code.jquery.com/jquery-3.1.0.js" >
  </head>
  <body>
    <div id='myID'>
      <h1 class="italic"> Hello World</h1>
      <p class="italic"> jQuery is Awesome!</p>
      <p> we can do so much </p>
      <p> like flip a coin: </p>
      <p id='result'> ... </p>
    </div>
  </body>
</html>
```



Hello World

jQuery is Awesome!

we can do so much

like flip a coin:

...

Challenge Time!

Use the code outline below to implement a coin flipping program

```
$( "element" ).action()
```

```
if ( Math.random() > 0.5 ) {  
    // change the text of #results to "heads"  
}  
else {  
    // change the text of #results to "tails"  
}
```

Code Along

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Students, write your response!

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jQuery: Key Points

— — —

- **jQuery** is a library that provides functions for manipulating HTML elements and CSS styles using JavaScript code.
- All jQuery code involves
 1. interacting with an **HTML element**
 2. performing an **action** using/on/with that element.

[Try it out!](#)

A rectangular button with a light gray background and a thin black border. The word "show" is written in black text on the button. A mouse cursor, represented by a small black arrow, is pointing at the bottom right corner of the button.

```
// General Syntax:  
$ ("element").action()  
  
// Example: hide an element  
$ ("p").hide()
```

jQuery: Element Creation

[Try it out!](#)

Create *new* HTML elements by including "<>":

```
$( "<elementType>" )
```

New elements created this way must be *appended* to an existing element.

```
<body>
  <div id="container"></div>
</body>

<script>
  $( "<h1>" ).appendTo( "#container" ).text( "hello" );
  $( "<p>" ).appendTo( "#container" )
    .text( "hello" )
    .css( "color", "red" );
</script>
```

Output

hello

hello

jQuery: Element Access

— — —

- **jQuery** can affect individual elements or groups of elements
- The extent of effect depends on the argument given.
- Arguments can be either selectors or tags.

```
// Affect all p elements:  
$("p").hide()  
  
// Affect all of class "text"  
$(".text").hide()  
  
// Affect only id "special"  
$("#special").hide()  
  
// Create a new <div> and  
// affect it immediately  
$("<div>").hide()
```

jQuery Events



jQuery can make websites interactive by responding to user inputs called **events**.

An event "fires" at the precise moment when something happens - like the user clicking on a button!

What are some other examples of "events" that might fire?

```
$("#button").on("click", toggleSecret);  
  
function toggleSecret() {  
    // code for showing/hiding  
    // the secret message  
}
```



Students, write your response!

jQuery Events and Event Handler Functions

— — —

The `.on("event", handlerFn)` function registers an **Event Handler** function to be called when an event fires.

```
$("button").on("click", toggleSecret);

// Event handler function
function toggleSecret() {
    // code for showing/hiding
    // the secret message
}
```

[Full Event List](#)

Mouse Events	Keyboard Events	Form Events	Window Events
click	keypress	submit	load
dblclick	keydown	change	resize
mouseenter	keyup	focus	scroll
mouseleave		blur	unload



Code Challenge

- 1) In the body, add a `<button>` element with the text "click me"
- 2) In the JavaScript, create a new function `coinFlip`
- 3) Register `coinFlip` as the eventHandler for a "click" event on the button.
- 4) Move all if/else statements into the `coinFlip` function.

```
$("#button").on("click", toggleSecret);  
  
// Event handler Function  
function toggleSecret() {  
    // code for showing/hiding  
    // the secret message  
}
```

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Students, write your response!

jQuery Events

— — —

Functions can be passed as input to **other functions** in the same way that other data values can.

The function `.on` is responsible for calling the function passed to it and is known as a *higher order function*.

No ()!



```
$("#button").on("click", coinFlip);
```

```
// Event handler Function  
function coinFlip() {  
    // code flipping a coin  
}
```

jQuery Events

Functions can be passed as input to other functions in the same way that other data values can.

The function `.on` is responsible for calling the function passed to it and is known as a *higher order function*.

No ()!

```
$("#button").on("click", coinFlip);  
  
// Event handler Function  
function coinFlip() {  
    // code flipping a coin  
}
```

What would happen if we wrote:

```
$("#button").on('click', coinFlip());
```



Students, write your response!

Some jQuery Tricks

[Try it out!](#)

Create *new* HTML elements by including "<>":

```
$("<elementType>")
```

New elements created this way must be *appended* to an existing element.

```
<body>
  <div id="container"></div>
</body>

<script>
  $("<h1>").appendTo("#container").text("hello");
  $("<p>").appendTo("#container")
    .text("hello")
    .css("color", "red");
</script>
```

Output

hello

hello

Some jQuery Tricks

[Try it out!](#)

— — —

Multiple actions can be *chained* together when they affect the same element.

Chained actions can be separated by new lines to improve readability.

```
<body>
  <div id="container"></div>
</body>

<script>
  $("<h1>").appendTo("#container").text("hello");
  $("<p>").appendTo("#container")
    .text("hello")
    .css("color", "red");
</script>
```

Output

hello

hello

Some jQuery Tricks

[Try it out!](#)

— — —

"jQueryes" can be saved in variables to avoid repeatedly querying for the same element. Variables holding jQuery elements should have names that begin with "\$"

```
<body>
  <div id="container"></div>
</body>

<script>
  var $container = $("#container");
  $("<h1>").appendTo($container).text("hello");
  $("<p>").appendTo($container)
    .text("hello")
    .css("color", "red");
</script>
```

Output

hello

hello

Some jQuery Tricks

— — —

"jQueryes" can have their attributes (including classes and ids) affected using jQuery. CSS can also be affected and retrieved using jQuery!

```
$myDiv = $("<div>");  
  
// Give $myDiv a class  
$myDiv.addClass("myClass");  
  
// Remove a class from $myDiv  
$myDiv.removeClass("myClass");  
  
// Give $myDiv an id  
$myDiv.attr("id", "myId");
```

```
$myDiv = $("<div>");  
  
// Give $myDiv blue text  
$myDiv.css("color", "blue");  
  
// Save the css for "color" in  
// a variable  
var color= $myDiv.css("color");
```

Code Challenge

Using only jQuery recreate the following HTML structure inside the existing #container element:

```
<div id="container">
  <h1> Favorite Ice Cream Flavors </h1>
  <ul>
    <li> Mint Chip </li>
    <li> Oreo </li>
    <li> Coffee </li>
  </ul>
</div>
```

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Students, write your response!

jQuery Reference

[\(REFERENCE LINK\)](#)

```
var $element = $("selector");           // references an existing element
var $otherElement = $(""); // creates a new element. don't forget the "<>"!

$element.appendTo($otherElement)        // inserts $element as a child of $otherElement
$element.detach()                       // removes the $element from its parent
$element.empty()                        // removes all children elements from $element

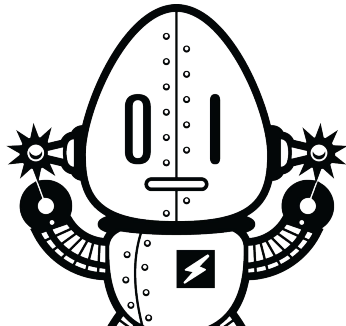
$element.css("left", 100);               // draws the $element offset 100 pixels from the left of the screen
$element.css("top", 50);                 // draws the $element offset 50 pixels from the top of the screen
$element.css("background-color", "red") // sets the background color of $element to "red"

$element.addClass("myClass");            // adds a class attribute for $element
$element.removeClass("myClass");         // removes a class attribute from $element
$element.attr("id", "abc")               // sets the id attribute value for $element
$element.attr("src", "img/file.png");    // sets the src attribute value for $element

$element.on("click", fn);                 // execute fn() in response to a "click" event
$element.on("keypress", fn);             // execute fn() in response to a "keypress" event
```

Quiz

ASD - jQuery



jQuery Mini Project: Dice App

1. Being creative requires you to understand the Fundamentals.
 - We'll practice HTML, CSS, JavaScript, and jQuery
2. We are never aiming to build the entire project from start to finish.
 - Code → Refactor → Code → Refactor
3. There is no one way to write a program
 - Try writing this program in a different way!